



ACTE — Dev Guide & Technical Specifications

This document accompanies the provided interactive HTML prototype. The prototype contains the complete UI/UX, responsive styles (Tailwind CSS), and the Light/Dark mode logic. Your role is to build the Backend infrastructure and the Desktop Capture Agent.

1. Project Overview & Architecture

ACTE is an automatic, privacy-focused time-tracking SaaS designed for lawyers.

To ensure strict professional secrecy, the application uses a **hybrid architecture**:

1. **The Desktop Capture Agent (Local):** A lightweight background app running on the lawyer's PC/Mac. It tracks the active window title (e.g., "*Word - Conclusions_Dupont.docx*") and measures active time locally.
2. **The Cloud SaaS (This Web App):** A centralized dashboard where lawyers review, edit, and validate their tracked activities, manage their law firm, and handle billing.

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[Local Desktop Agent] —(Encrypted API Sync)—> [SaaS Backend (Node/Go/Python)] —> [HTML Dashboard]
```

2. Your Tech Stack Objectives

A. The Desktop Capture Agent (The "Sonde")

- **Recommended Technology:** Electron or Rust/Tauri (Tauri is highly preferred for its minimal footprint and security).
- **Core Features:**
 - Must run silently in the background (system tray icon).
 - Request OS accessibility permissions to read active window titles.
 - Local configuration screen: user inputs their **Personal API Key** (retrieved from the SaaS Web App profile page).
 - **Privacy Filter (Local processing):** Apply local regex or simple classification models to only extract *Duration* (e.g., **24 min**) and *Metadata* (e.g., document name/type), without ever uploading the full document content.
 - Encrypt data before sending it via REST API.

B. The Web App Backend (SaaS API)

- **Recommended Technology:** Node.js (Express/NestJS) or Go.
- **Database:** PostgreSQL.
- **Core APIs to build:**
 1. **Authentication & Subscription:** Standard user auth + Stripe integration for the organization subscription (50€/user/month).
 2. **Agent Data Ingest:** Endpoint for the Desktop Agent to push encrypted time logs.
 3. **Dashboard Data Query:** Serve aggregate data for the "Journal" page and "Admin Console" page.

3. Mapping Front-End to Back-End (Interactive Elements)

1 The Desktop Agent Download

- **UI Location:** Sidebar bottom left (`#download-agent-btn`).
- **Expected Dev Action:** Host the compiled `.dmg` (Mac) and `.exe` (Windows) binary files of the Desktop Agent on a secure bucket (S3) and link this button to initiate the download.

2 Personal API Key (User Activation)

- **UI Location:** "Mon Profil" Tab (`#api-key-container`).
- **Expected Dev Action:**
 - Generate a unique, secure API Key (`acte_live_...`) for every user upon registration.
 - Securely store it (hashed) in the database.
 - The web page must let the user copy this key so they can paste it into their downloaded Desktop Agent to link their local tracking activity.

3 Stripe Integration (Subscription Management)

- **UI Location:** Admin Console -> "Abonnement Cabinet" tab (`#manage-stripe-btn`).
- **Expected Dev Action:**
 - Connect Stripe Webhooks.
 - Clicking "Gérer le paiement" should redirect the admin to the **Stripe Customer Portal** where they can update credit cards, download invoices, or scale user licenses (e.g., 250€/month for 5 active seats).

4 Admin Actions (Firm Management)

- **UI Location:** Admin Console -> Team Table.
 - **+ Inviter un membre:** Must open the invitation modal, send an invitation email, and add the user with `Pending` status to the DB.
 - **Action Button (⋮):**
 - *Modifier le taux:* Update the hourly rate of the collaborator (e.g., 200€/h) in the database.
 - *Rappeler la validation:* Trigger a transactional email reminder via SendGrid/Postmark to the collaborator who has low validation rates (e.g., Me Okafor with 89% validation).
 - *Suspendre le compte:* Set database user status to `Inactive` and instantly revoke their session / API Key accesses.

4. Strict Security Guidelines (Non-negotiable)

- **Zero-Knowledge Principle:** We are dealing with lawyers bound by attorney-client privilege. The database must **never** store raw, unencrypted document text.
- **Separation of Environments:**
 - **Staging:** You will work on your own staging servers with mocked, non-real client data.
 - **Production:** Hosted on European sovereign clouds (e.g., OVHcloud or Scaleway) with restricted, encrypted SSH keys. Only the core founders hold the production keys.